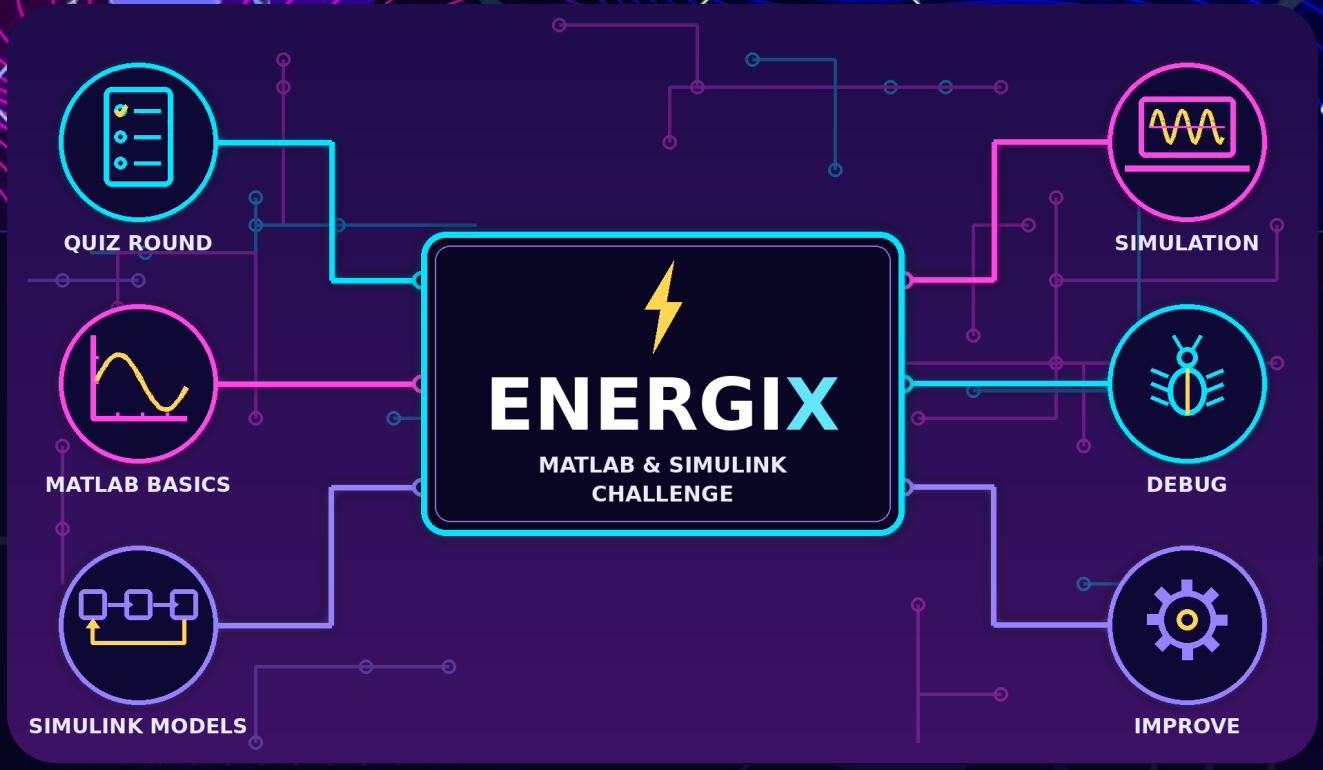
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OFFICIAL EVENT RULEBOOK

1. About the Event

ENERGIX is a technical competition designed to evaluate participants' knowledge of MATLAB and Simulink, their circuit simulation skills, and their ability to troubleshoot and improve electrical circuit models. The event consists of three rounds that progress from fundamental theoretical knowledge to practical simulation and circuit debugging.

2. Team Formation

- Each team must consist of a **minimum of 2 members and a maximum of 3 members**.
- Participants must register their teams according to the instructions provided by the organizers.
- All team members are expected to contribute to the assigned tasks throughout the competition.

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3. Event Structure and Mark Distribution

The competition consists of three rounds, with a total of 100 marks.



Round	Name	Maximum Marks
Round 1	Quiz Round	**30**
Round 2	Simulation Round	**30**
Round 3	Debug and Improve a Circuit Round	**40**
Total		**100**

4. Round 1: Quiz Round — 30 Marks

Objective

To assess participants' fundamental knowledge of MATLAB and introductory Simulink concepts.

Topics Covered

- MATLAB fundamentals, commands, and syntax.
- Variables, operators, and basic programming concepts.
- Introduction to Simulink blocks and their applications.

Rules

1. Participants will answer questions based on the topics listed above.
2. The round will carry a maximum of **30 marks**.
3. Participants must complete the quiz within the time limit specified by the organizers.
4. The use of unauthorized materials, external assistance, or communication with other teams during the quiz is **prohibited**.

5. Round 2: Simulation Round — 30 Marks

Objective

To evaluate participants' ability to model, simulate, and analyze electrical circuits using MATLAB/ Simulink.

Procedure

1. Each team will receive a circuit problem statement with specified input parameters and a target output.
2. The challenge may involve an electronics circuit.
3. Teams must construct or configure the circuit model in MATLAB/Simulink.
4. The final simulation model and output waveform must be submitted to the organizers for evaluation.

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Rules and Requirements

- Each team must bring a **laptop with MATLAB and Simulink installed and tested before the event**.
- Participants are strongly encouraged to install **MATLAB R2025a**, if possible, to minimize software compatibility issues.
- Teams are responsible for ensuring that their software is functional before the competition begins.
- All circuit parameters and target output requirements necessary to solve the problem will be provided in the problem statement.
- The simulation must be completed within the time limit announced by the organizers.
- Marks will be awarded based on the simulation process, parameter optimization, output accuracy, waveform verification, and submission of the completed model.

Maximum Marks: 30

6. Round 3: Debug and Improve a Circuit Round — 40 Marks

Objective

To assess participants' ability to identify faults, troubleshoot errors, improve circuit models, and verify simulation results using MATLAB/Simulink.

Procedure

1. Each team will receive a Simulink circuit model containing deliberately introduced faults.
2. Teams must analyze the circuit and identify the errors in the model.
3. Participants must correct the identified faults and make the necessary improvements to the circuit.
4. Teams must run the simulation successfully and achieve the specified output.
5. The corrected simulation model and output waveform must be submitted for evaluation.
6. Teams must explain their debugging approach and the corrections made to the judges when requested.

Rules

- Teams must work on the circuit model provided by the organizers.
- Participants must identify and correct the faults independently without unauthorized assistance.
- Teams must ensure that their final model runs successfully and meets the specified output requirements.
- Marks will be awarded based on fault identification, accuracy of corrections, output performance, waveform verification, and technical explanation.

7. General Rules and Regulations

1. All participants must follow the instructions provided by the event organizers.
2. Teams must complete each round within the allotted time.
3. Participants must maintain discipline and demonstrate fair play throughout the competition.
4. Copying another team's work, sharing solutions during a round, or using unauthorized assistance is **prohibited**.
5. Teams must submit their work in the format and manner specified by the organizers.
6. The organizers reserve the right to clarify rules and resolve unforeseen issues to ensure fair and orderly conduct of the event.

8. Final Evaluation

- The maximum score for the competition is **100 marks**.
- Final rankings will be determined by the **aggregate marks** obtained across all three rounds.
- In the event of a tie, the team with the **Lowest time in all Round** will be given preference.

- The judges' decisions regarding technical evaluation and final rankings will be considered **final**.

9. Prize Pool

Prize Value: ₹17,000+ (including goodies)

Registration Charge

REGISTRATION FEE

₹ 50 per member

Contact Details

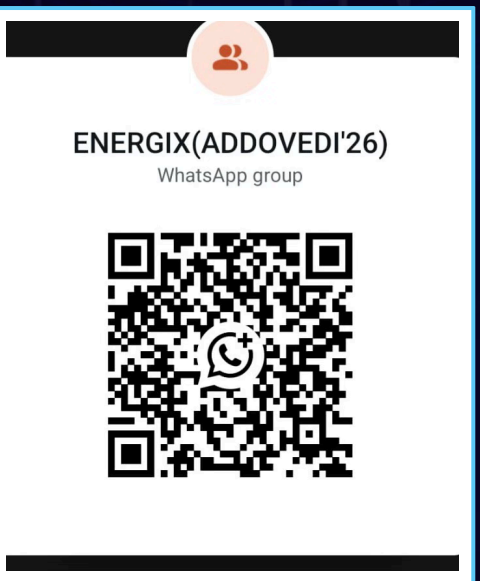
EVENT COORDINATORS

Prakash Tamang

Roshni Yadav

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Join our WhatsApp group for all the future updates!



Best Wishes to All Participants!

ENERGIX — Think. Simulate. Debug. Innovate.

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